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ORIGINAL ARTICLE

Fasxiator, a novel factor XIa inhibitor from snake venom, and its site-specific mutagenesis to improve potency and selectivity

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Summary. *Background:* Bleeding remains a major limitation of standard anticoagulant drugs that target the extrinsic and common coagulation pathways. Recently, intrinsic coagulation factors are increasingly being investigated as alternative targets for developing anticoagulant drugs with lower bleeding risk. *Objectives:* Goals were to (i) identify novel anticoagulants selectively targeting intrinsic coagulation pathway and (ii) characterize and further improve the properties of the identified anticoagulants. *Methods and Results:* We have isolated and sequenced a specific factor XIa (FXIa) inhibitor, henceforth named Fasxiator, from the venom of the banded krait snake, *Bungarus fasciatus*. It is a Kunitz-type protease inhibitor that prolonged activated partial thromboplastin time without significant effects on prothrombin time. Fasxiator was recombinantly expressed (rFasxiator), purified, and characterized to be a slow-type inhibitor of FXIa that exerts its anticoagulant activities (doubled activated partial thromboplastin time at ~ 3 μmol L^{-1}) by selectively inhibiting human FXIa in *in vitro* assays. A series of mutants were subsequently generated to improve the potency and selectivity of recombinant rFasxiator. $rFasxiator_{N17R,L19E}$ showed the best balance between potency (IC_{50} ~ 1 nmol L^{-1}) and selectivity (> 100 times). $rFasxiator_{N17R,L19E}$ is a competitive slow-type inhibitor of FXIa (K_i = 0.86 nmol L^{-1}), possesses anticoagulant activity that is ~ 10 times stronger in human plasma than in murine plasma, and prolonged the occlusion time of mice carotid artery in $FeCl_3$-induced thrombosis models. *Conclusion:* We have isolated an exogenous FXIa specific inhibitor, engineered it to improve its potency by ~ 1000 times and demonstrated its *in vitro* and *in vivo* efficacy. These proof-of-principle data supported the further development of Fasxiator as a novel anticoagulant candidate.

Keywords: anticoagulants; carotid artery thrombosis; factor XIa; protease inhibitors; snake venom.

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Introduction

Blood coagulation is an important physiological process to prevent excessive blood loss during injury [1]. The blood coagulation cascade can be divided into three pathways: intrinsic, extrinsic, and common pathways. Factor Xa (FXa), thrombin, Factor VIIa (FVIIa), and tissue factor participating in the common and extrinsic pathways are prime targets for designing antithrombotic drugs, as the inhibition of these proteins can easily inhibit clot formation [2]. However, such drugs are usually associated with high risks of excessive bleeding [3,4]. Therefore, strategies that inhibit abnormal thrombus formation without impeding the clotting response to tissue injuries are highly attractive. One such strategy would be to target components along the intrinsic pathway which will slow the coagulation cascade in the event of intravascular thrombosis, yet not result in aberration to clot formation in response to injury. This theory is supported by the observation that patients with deficiencies of proteins along the intrinsic pathway exhibit significantly milder bleeding symptoms compared with patients deficient in factors of the extrinsic or common pathways [5].

Factor XI (FXI) is an important serine protease zymogen in the intrinsic pathway. It is suggested to play

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a significant role in animal thrombosis models [6,7]. In contrast to FX and thrombin, disruption of FXI resulted in little bleeding tendency [8]. Further, patients with deficiency in plasma FXI experienced a lower probability of deep vein thrombosis and ischemic stroke [9,10]. Thus, FXI has been recognized in recent years as a new drug target for antithrombotic drug design.

Snake venoms represent a prime source of novel drug leads for the development of new anticoagulants. We have been interested in snake venom proteins that target the hemostatic system to induce procoagulant or anticoagulant effects [11,12]. In this report, we describe the isolation, recombinant expression, and structural and functional characterization of a novel Kunitz-type protease inhibitor from *Bungarus fasciatus* (banded krait) venom that selectively inhibits FXIa (henceforth termed Fasxiator: *Bungarus fasciatus* FXIa inhibitor) while exhibiting negligible inhibition on other proteases of the blood coagulation cascade. Further, through systematic and selective point mutations, we have designed a series of variants in bid to create a highly potent and specific FXIa inhibitor. The most promising mutant, $Fasxiator_{N17R,L19E}$, was evaluated for *in vitro* inhibitory kinetics and *in vivo* efficacy in a thrombotic model.

Methods

Effect on activated partial thromboplastin time

Equal volumes (50 μL) of human plasma, APTT-XL (Pacific Hemostasis, Thermo Scientific, Waltham, MA, USA), 20 mmol L^{-1} $CaCl_2$, and various concentrations of inhibitor diluted with HEPES buffer (25 mmol L^{-1}, pH 7.4) were prewarmed to 37 °C individually. Inhibitor was added to plasma and incubated for 10 min, followed by the addition of APTT-XL and further incubation for 3 min. Clotting was initiated by the addition of 20 mmol L^{-1} $CaCl_2$ to the mixture, and the clot formation was measured using an Infinite M200 microplate reader [13] (Tecan, Männedorf, Switzerland) at 650 nm for fibrin polymer formation [14].

Prothrombin time

Plasma, thromboplastin-D (Pacific Hemostasis), and inhibitor diluted with HEPES buffer were prewarmed to 37 °C individually. Inhibitor (100 μL) was added to plasma (100 μL) and incubated for 10 min. Clotting was initiated by the addition of thromboplastin-D (100 μL) to the mixture, and the clot formation was measured using a BBL fibrometer (Becton Dickinson and Co., Sparks, MD, USA) [15].

Protease selectivity profile

Respective serine protease (50 μL) diluted with buffer (50 mmol L^{-1} HEPES, pH 7.4, 140 mmol L^{-1} NaCl, 5 mmol L^{-1} $CaCl_2$, 0.1% BSA) was preincubated with 25 μL of inhibitor for 30 min at 37 °C, followed by the addition of 25 μL of the appropriate chromogenic substrate. In a total volume of 100 μL, the final serine protease/substrate concentrations were as follows: FVIIa (200 nmol L^{-1})/S-2288 (1 mmol L^{-1}), FIXa (150 nmol L^{-1})/Spectrozyme FIXa (1 mmol L^{-1}), FXa (2 nmol L^{-1})/S-2222 (0.5 mmol L^{-1}), FXIa (0.5 nmol L^{-1})/S-2366 (0.5 mmol L^{-1}), FXIIa (10 nmol L^{-1})/S-2302 (0.5 mmol L^{-1}), kallikrein (16 nmol L^{-1})/S-2302 (0.5 mmol L^{-1}), thrombin (2 nmol L^{-1})/S-2238 (0.1 mmol L^{-1}), activated protein C (APC) (5 nmol L^{-1})/S-2366 (1 mmol L^{-1}), urokinase (400 ng mL^{-1})/S-2444 (0.5 mmol L^{-1}), plasmin (5 nmol L^{-1})/S-2251 (0.5 mmol L^{-1}), and chymotrypsin (10 nmol L^{-1})/S-2586 (0.5 mmol L^{-1}). The cleavage of substrate was measured at 405 nm using a microplate reader.

Because the concentrations of rFasxiator variants (1 nmol L^{-1} to 10 μmol L^{-1}) used were much lower than those of rFasxiator (30–120 μmol L^{-1}), the concentration of the serine proteases were concomitantly decreased (100 nmol L^{-1} FVIIa, 1 nmol L^{-1} FXa, 5 nmol L^{-1} FXIIa, 2 nmol L^{-1} kallikrein, 1 nmol L^{-1} thrombin, 2 nmol L^{-1} APC, 1 nmol L^{-1} plasmin, 5 nmol L^{-1} chymotrypsin) to increase the ratio of inhibitor to enzyme. The substrate concentrations of FXa (1 mmol L^{-1} S2222), FXIa (1 mmol L^{-1} S2366), and APC (1.5 mmol L^{-1} S2366) were increased to allow for a longer monitor time.

Kinetic studies

The kinetic study was carried out according to the methods described by Copeland [16].

K_i determination for slow-type inhibitor The progress curves were fit into Eqn. 1 using KaleidaGraph 4.5 software, Synergy Software, Reading, PA, USA [17].

$$P = V_s t + (V_i - V_s)\,[1 - \exp(-K_{obs} \times t)]/K_{obs} + P_0 \quad (1)$$

P is the amount of product, V_s is the steady state velocity, V_i is the initial state velocity, t is the time, P_0 is the correction factor for the initial amount of product, and K_{obs} is the apparent first-order rate constant for the interconversion between V_s and V_i.

Obtained K_{obs} values were plotted against substrate concentration and inhibitor concentration.

For type I slow-type inhibitor [16], the relationship between K_{obs} and inhibitor concentration [I] is given by Eqn. 2:

$$K_{obs} = K_4(1 + [I]/K_i^{app}) \quad (2)$$

For competitive inhibitor, the relationship between K_i^{app} and K_i was given by Eqn. 3:

$$K_i = K_i^{app}/(1 + [S]/K_m) \quad (3)$$

where $[S]$ is substrate concentration and K_m is the Michaelis constant.

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K_i determination for fast-type inhibitor For fast-type inhibitors, their K_i values were calculated with Eqn. 4 [17].

$$K_i = IC_{50}/(1 + [S]/K_m) \quad (4)$$

IC_{50} values were determined by fitting the dose-inhibition curve using GraphPad Prism software (San Diego, CA, USA).

$FeCl_3$-induced carotid artery thrombosis model

All animal experiments were carried out under protocol 041/12 approved by Institutional Animal Care and Use Committee, National University of Singapore. The $FeCl_3$-induced carotid artery thrombosis model was performed as described previously [18–20]. C57BL/6 male mice (9–11 weeks old, 24.5–27.5 g) were anesthetized with an intraperitoneal injection of ketamine (75 mg kg^{-1}) and medetomidine (1 mg kg^{-1}). Two sets of independent experiments were performed to evaluate the efficacy of $rFasxiator_{N17R,L19E}$. In the first experiment, 0.1 mL of inhibitor (2 mg mL^{-1}) was injected into the mice via the tail vein. The right carotid artery was exposed using blunt dissection, and vascular injuries were caused by applying filter paper of 2 mm × 2 mm saturated with 6% $FeCl_3$ on top of the carotid artery for 3 min. In the second experiment, a greater amount of inhibitor (0.15 mL) was injected, and intensity of injury was decreased by using smaller filter paper (1 mm × 1 mm) and lower $FeCl_3$ (5%) concentration. After 3-min $FeCl_3$ exposure, the filter paper was removed and the vessel was washed with sterile normal saline. To determine the time to occlusion, a miniature Doppler flow probe (catalog No. MA0.5VB; Transonic Systems Inc., Ithaca, NY, USA) was placed around the carotid artery, and blood flow was recorded using a Transonic flowmeter. The maximal monitor time after injury was 30 min. Mice were killed via cervical dislocation immediately after conclusion of the experiment and prior to recovery from anesthesia.

Statistical analysis

Results are expressed as means ± SD unless otherwise described. Statistical differences among the animal groups were analyzed by unpaired *t*-test. Significance was set at $P \leq 0.05$.

Results

Isolation of anticoagulants that selectively target intrinsic pathway

A plethora of different venom toxins, including three-finger toxins, Kunitz-type protease inhibitors, natriuretic peptides, and C-type lectins, have been isolated from the krait (*Bungarus* sp.) venoms. To isolate anticoagulant proteins, we subjected *B. fasciatus* venom to a two-step fractionation using size exclusion chromatography and reverse-phase high-performance liquid chromatography (RP-HPLC) (Fig. 1A–F). The molecular weight (M_r) of each RP-HPLC fraction was then determined by electrospray ionization–mass spectrometry (ESI-MS). Pooled fractions containing proteins < 8 kDa (Table S1) were tested for their effects on activated partial thromboplastin time (APTT) and prothrombin time (PT) (Fig. 1G).

Three RP-HPLC pools (B1, C1, and D1) prolonged APTT without significant effects on PT (Fig. 1G), indicating that they affected the intrinsic pathway, while leaving the extrinsic pathway unaffected. ESI-MS results indicated that these pools mainly contained two proteins with M_r of 6977.5 ± 0.85 Da and 7307.3 ± 0.67 Da. These two proteins were separated by introducing a cation-exchange chromatography step between size exclusion chromatography and RP-HPLC (Fig. 1H–J). The homogeneity and mass of the purified proteins were determined by ESI-MS. The smaller molecule (M_r = 6977.5 ± 0.85 Da) was named BF01, and the larger molecule (M_r = 7307.3 ± 0.67 Da) was named BF02. Both BF01 and BF02 prolonged APTT (Fig. 1K).

Fig. 1. Identification of novel anticoagulants from *Bungarus fasciatus* venom. (A) Size exclusion chromatography of *B. fasciatus* venom. Crude venom (100 mg, a generous gift from Professor Cassian Bon, Pasteur Institute, Paris, France) was dissolved in 50 mmol L^{-1} Tris-HCl buffer (pH 8.0) and separated on a HiLoad 16/60 Superdex 30 preparative grade column. Fractions (1 mL) were collected and pooled into pools A–E. Individual pools were applied to a Jupiter C18 (250 × 10 mm) reversed-phase column and resolved by a multistep gradient elution profile (flow rate 5 mL min^{-1}) with buffer A being 0.1% TFA in water and buffer B being 0.1% TFA in 80% acetonitrile water (B–F). Fractions with M_r < 8 kDa (Table S1) were pooled into A1, A2, A3, A4, B1, B2, C1, D1, D2, E1, and E2. (G) Anticoagulant activities of RP-HPLC fractions. Pooled fractions were evaluated using APTT and PT in triplicate. Error bars represent standard deviations. Pools B1, C1, and D1 prolonged APTT without significant effects on PT and were further characterized. MS studies revealed that pools B1, C1, and D1 mainly contain two proteins with M_r of 6977.5 ± 0.85 and 7307.3 ± 0.67 Da (Table S1). (H) Pooled size exclusion chromatography fractions containing anticoagulant proteins were loaded on a Bio-Rad S6 cation exchange chromatography column. Peak 1 contained BF01 (M_r = 6977.5 ± 0.85 Da), and peak 2 contained BF02 (M_r = 7307.3 ± 0.67 Da). (I and J) BF01 and BF02 were further fractionated using RP-HPLC. Fractions containing BF01 and BF02 were pooled and lyophilized for future use. (K) BF01 and BF02 prolong APTT at a final concentration of 3.75 μmol L^{-1}. (L) Effects of BF01 on various procoagulant proteases in the coagulation cascade were evaluated. Results indicate that both proteins are selective inhibitors of FXIa. (M) Alignment of the protein sequences with relevant Kunitz-type inhibitors. Dark gray: cysteines; light grey: loop 1 and loop 2 residues. The disulfide bridges were shown by solid lines.

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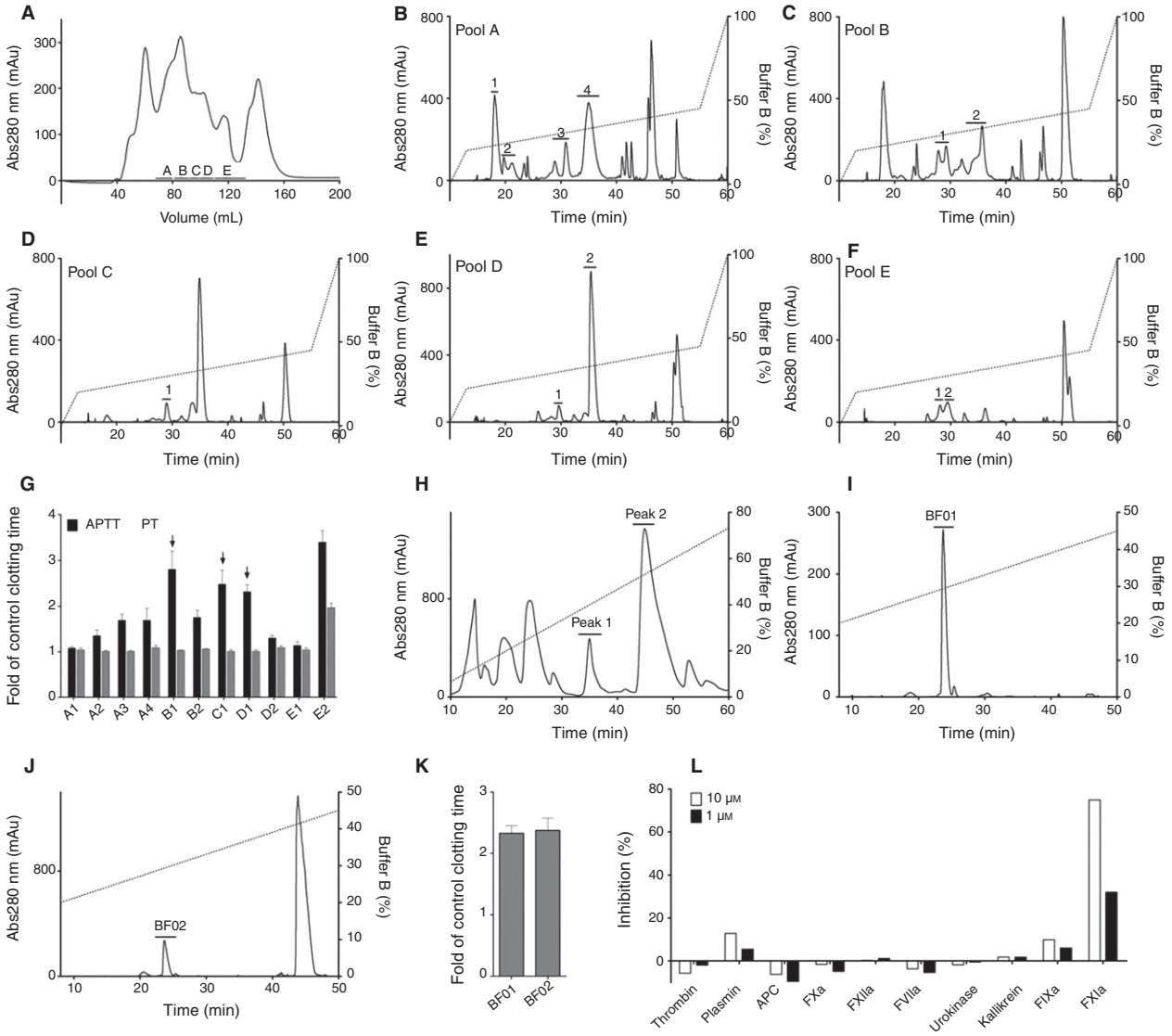
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Protease specificity of novel anticoagulants

To determine the protease specificity, BF01 and BF02 were independently screened against several procoagulant serine proteases in the coagulation pathway (Fig. S1) and were found to selectively inhibit FXIa. The more abundant BF01 was further screened against 10 serine proteases (thrombin, FVIIa, FIXa, FXa, FXIa, FXIIa, plasmin, kallikrein, APC, urokniase) in the coagulation cascade. Other than FXIa, BF01 (10 μmol L^{-1}) did not inhibit the activities of other serine proteases (Fig. 1L).

Amino acid sequences of novel anticoagulants

BF01 and BF02 were subjected to pyridylethylation, Lys-C/Arg-C digestion, and protein sequencing. Elution profile of peptide fragments and sequencing data suggested that BF01 and BF02 are isoforms (Fig. S2): BF02



M

Protein Name	Amino acid sequences	Identities (%)
BF01	KNRPTFCNLLPETGRCNALIPAFYYNSHLRKCQKFNYGGCGGNANNFKTIDECQRTCAAKYG---	100
BF02	KNRPTFCNLLPETGRCNALIPAFYYNSHLRKCQKFNYGGCGGNANNFKTIDECQRTCAAKYGRSS	100
BFIX	KNRPTFCNLLPETGRCNALIPAFYYNSHLHKCQKFNYGGCGGNANNFKTIDECQRTCAAKYGRSS	98
PN2KPI	----EVCSEQAETGPCRAMISRWYFDVTEGKCAPFFYGGCGGNRNNFDTEEYCMAVCGSAI----	45
BPTI	--RPDFCLEPPYTGPCKARIIRYFYNAKAGLCQTFVYGGCRAKRNNFKSAEDCMRTCGGA-----	49
HlChl	QRKPSLCYLPKDSGVCYAFFPSFYYDSATRTCRRFIYGGCMGNENRFRSFEECTSVCG-------	43
AvKTI	---KDRCLLPKVTGPCKASLTRYYYDKDTKACVEFIYGGCRGNRNNFKQKDECEKACTDH-----	49
BBPTI	HDRPKFCYLPADPGECLAHMRSFYYDSESKKCKEFIYGGCHGNANKFPSRDKCRQTCGGK-----	49

Disulfide bridges

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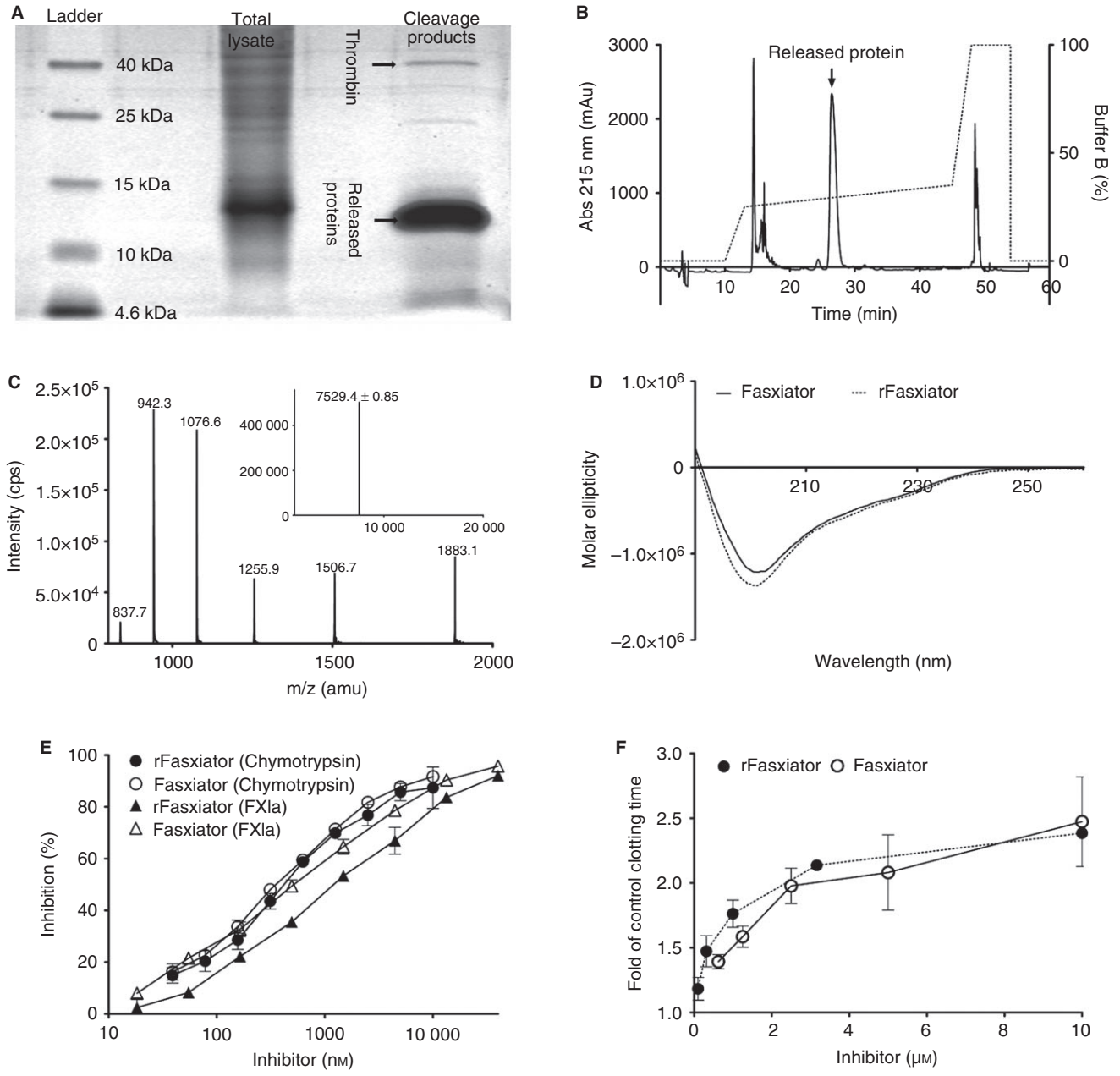


Fig. 2. Recombinant expression and purification of rFasxiator. (A) SDS PAGE of total bacterial lysate and thrombin cleavage product from Ni-NTA beads. The Fasxiator gene was synthesized according to the protein sequence and using *E. coli* preferred codons and then was cloned into pET28a expression vector between *Eco*RI and *Hind*III restriction sites. (B) Purification of thrombin cleavage products using RP-HPLC. (C) ESI-MS profile of purified recombinant protein (rFasxiator). rFasxiator has an additional five amino acid residues in the N-terminal compared with Fasxiator. (D) Circular dichroism spectra of Fasxiator and rFasxiator. Far-UV circular dichroism spectra (260–190 nmol L^{-1}) of protein MilliQ water solution (0.08–0.1 mg mL^{-1}) was recorded using a Jasco J-810 Spectropolarimeter. (E) Inhibition of chymotrypsin and FXIa activities by Fasxiator and rFasxiator. (F) Prolongation of APTT by Fasxiator and rFasxiator. Results indicate that Fasxiator and rFasxiator share similar structure and activity (D–F).

has three additional amino acid residues (RSS) at the C-terminal compared with BF01. BLAST searches indicated that BF01 and BF02 were Kunitz-type protease inhibitors. BF02 is only one residue different from a previously reported chymotrypsin inhibitor, Kunitz IX (BFIX), isolated from the venom of the same species [21]. However, its hemostatic effect has not been previously reported.

The sequences of BF01 and BF02 were aligned with BFIX, PN2KPI, and other Kunitz-type chymotrypsin or trypsin inhibitors (Fig. 1M). PN2KPI is the recombinant truncated Kunitz domain of protease nexin 2 (PN2). PN2 is a 120-kDa isoform of the Alzheimer β-amyloid protein precursor that has been shown to be a highly potent and physiologically relevant inhibitor of FXIa [22,23]. It is a slow, tight binding inhibitor of FXIa (K_i = 290–450 pmol

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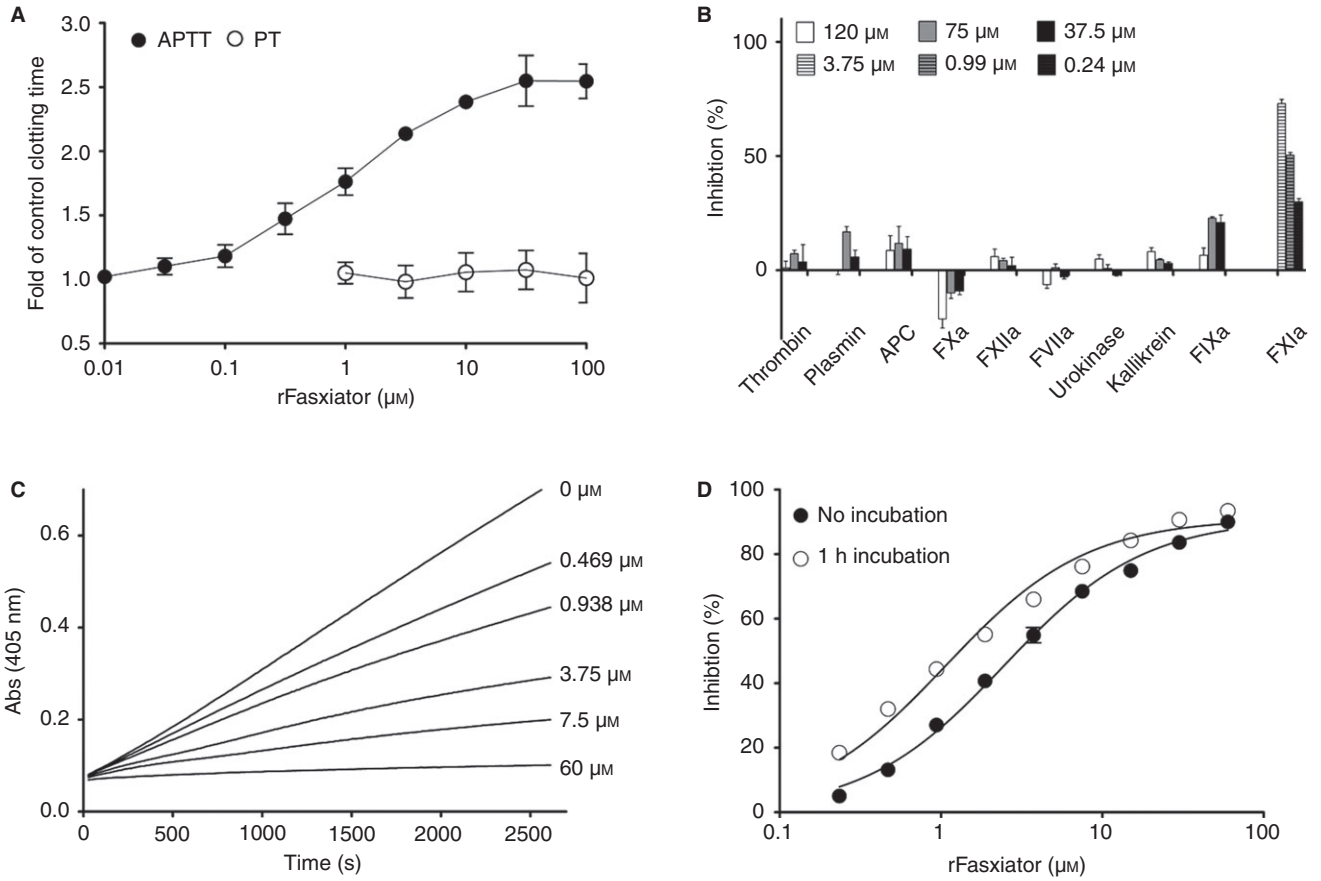


Fig. 3. Anticoagulant activity and protease specificity of rFasxiator. (A) Effect of rFasxiator on APTT and PT. rFasxiator prolonged APTT in a dose-dependent manner, while it had no significant effect on PT. (B) rFasxiator selectively inhibits FXIa among the serine proteases in the blood coagulation cascade. (C) Progress curves of S2366 cleavage by FXIa in the presence of different concentrations of rFasxiator. (D) Effects of incubation time on the inhibition of FXIa by rFasxiator. Dose-inhibition curve shifts to the left with 1-h incubation. Time-dependent increase in inhibitory effect of rFasxiator indicates that rFasxiator is a slow-type inhibitor.

L^{-1}), with little or no detectable inhibitory activity against other coagulation enzymes [22,23]. PN2KPI retains the inhibitory activity against FXIa in *in vitro* assays and is being evaluated as an antithrombotic agent [17,18]. Interestingly, BF01/BF02 shows limited similarity with PN2KPI (45% identity; 58% similarity). We focused on BF01 for further studies, and it was named as Fasxiator (*B. fasciatus* FXIa inhibitor).

Recombinant expression of Fasxiator

ESI-MS and protein sequencing by Edman degradation (Fig. S2) suggested that there is no post-translational modification in Fasxiator that requires eukaryotic expression system. Therefore, we used the *E. coli* system, which has been used for the successful expression of snake venom derived Kunitz-type inhibitors [24]. Further, it is the most robust and cheapest method with high yields for protein. Fasxiator was recombinantly expressed in *E. coli* using pET28a vector, purified, and refolded on Ni-NTA beads (Fig. 2A). The refolded protein was purified by RP-HPLC (Fig. 2B). rFasxiator has an additional five amino acid residues 'GSEFM' in the N-terminal compared with BF01. The identity and homogeneity of rFasxiator were ascertained to be 7529.40 ± 0.85 Da by ESI-MS (Fig. 2C), which matches the calculated mass of the sequence. rFasxiator showed similar secondary structure (Fig. 2D), inhibitory effects on FXIa and chymotrypsin (Fig. 2E), and anticoagulant activity (Fig. 2F) compared with native Fasxiator. Thus, rFasxiator is structurally and functionally similar to native Fasxiator and was used in subsequent functional studies.

rFasxiator selectively inhibits FXIa

The anticoagulant activity and protease specificity of rFasxiator were determined. rFasxiator doubled APTT at ~ 3 μmol L^{-1} but had no significant effect on PT up to 100 μmol L^{-1} (Fig. 3A). rFasxiator was tested for its effect on extrinsic and intrinsic tenase complexes (Fig. S3), as well as various serine proteases involved in the coagulation cascade (Fig. 3B). rFasxiator selectively inhibits FXIa (IC_{50} ~ 1.5 μmol L^{-1}); it failed to significantly inhibit other serine proteases even at 120 μmol L^{-}

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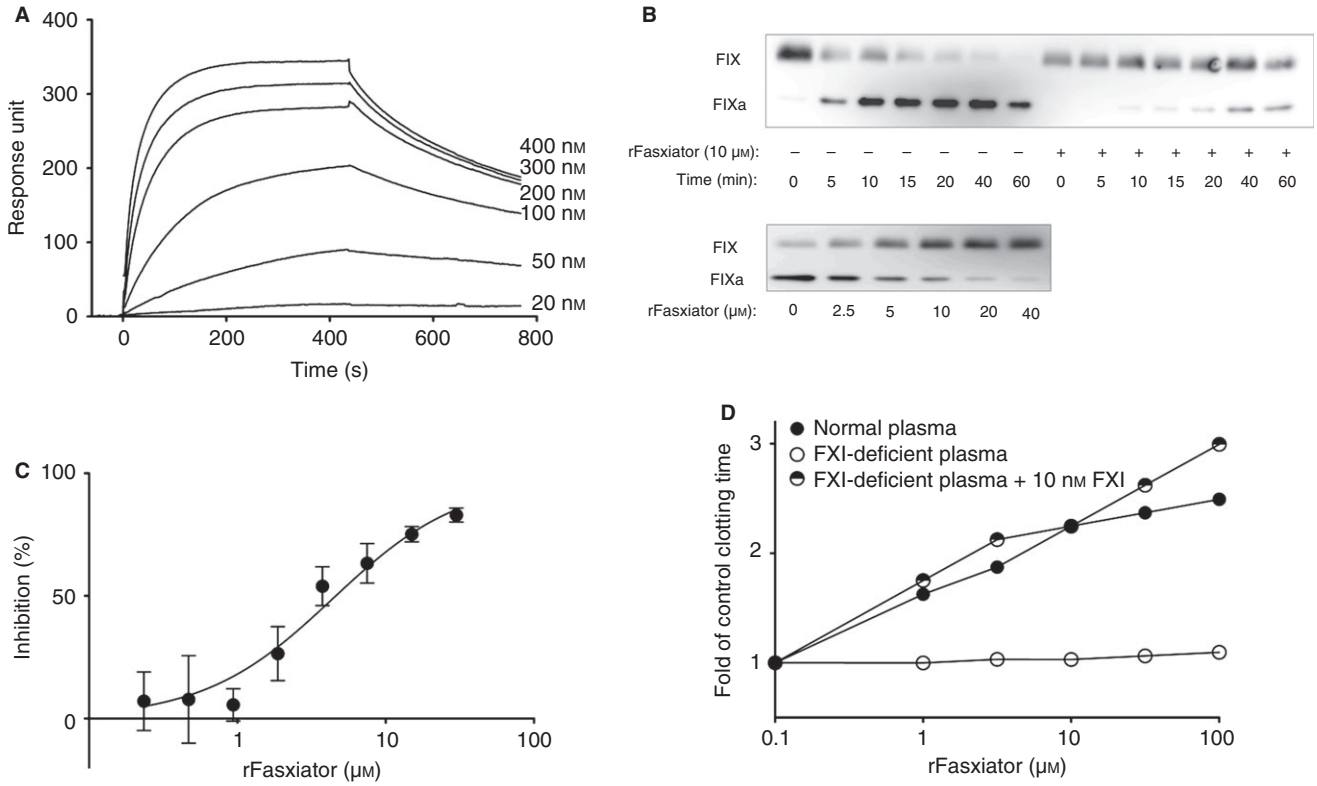


Fig. 4. rFasxiator interacts with and inhibits FXIa. (A) High affinity binding of rFasxiator to FXIa. In SPR studies, rFasxiator (50 μL of 500 nmol L^{-1}, in 10 mmol L^{-1} acetate buffer, pH 6.0) was immobilized on a SensiQ COOH5 chip using amine coupling with a SensiQ® Pioneer system. Then, 145 μL of different concentrations of FXIa or chymotrypsin diluted in running buffer (10 mmol L^{-1} HEPEs, 150 mmol L^{-1} NaCl, 5 mmol L^{-1} $CaCl_2$, and 0.1% Tween 20, pH 7.4) were used as analytes at a flow rate of 20 μL min^{-1}, and a dissociation time of 240 s was applied. The chip surface was regenerated with 25 μL of 10 mmol L^{-1} glycine/HCl buffer, pH 3.1, at a flow rate of 50 μL min^{-1} and a dissociation time of 240 s after each analysis cycle. FXIa binds to rFasxiator with a K_D of 20.2 nmol L^{-1}. rFasxiator also binds chymotrypsin with a K_D of 29.3 nmol L^{-1} (Fig. S4). (B) Inhibition of cleavage of FIX by rFasxiator. FIX and FIXa were detected by Western blotting. rFasxiator shows time-dependent (upper panel) and dose-dependent (lower panel) inhibition of activation of FIX. In the dosage-dependent assay: 5 μL of 750 pmol L^{-1} FXIa, 5 μL of 300 nmol L^{-1} FIX, and 5 μL of various concentration of recombinant inhibitor prewarmed to 37 °C were mixed and further incubated at 37 °C for 15 min before mixing with 5 μL of 4× SDS sample buffer and heated at 100 °C for 10 min to stop the reaction. In the time-dependent assay: 40 μL of 750 pmol L^{-1} FXIa, 40 μL of 300 nmol L^{-1} FIX, and 40 μL of buffer or 30 μmol L^{-1} recombinant proteins prewarmed to 37 °C were mixed. From this mixture, 15-μL aliquots were removed at indicated times, mixed with 5 μL of 4× SDS sample buffer, and heated at 100 °C for 10 min to stop the reaction. In both cases, all the 20 μL was then loaded to a 10% SDS-PAGE, transferred to a nitrocellulose membrane, and probed for FIX and FIXa using mouse anti-FIX antibody. (C) Inhibition of activation of FIX by rFasxiator. The FIXa generated was quantified by its hydrolytic activity on chromogenic substrate (activity in the absence of rFasxiator was defined as 100%) as described earlier [41]. (D) FXIa is the target for rFasxiator for its anticoagulant effects. APTT is prolonged by rFasxiator in normal plasma but not in FXI-deficient plasma. The anticoagulant effects were observed when 10 nmol L^{-1} FXI was added to FXI-deficient plasma. Data points represent average of duplicates. Thus, FXI is the main target of rFasxiator.

[1]. The protease selectivity profile (Figs 1L and 3B) showed that Fasxiator had no significant effects on FXIIa or thrombin (< 10% inhibition at 120 μmol L^{-1} compared with > 80% inhibition of FXIa at 3.75 μmol L^{-1}). Further, it was a competitive inhibitor (see later) and, thus, binds to the active site and not to exosite of FXIa. Therefore, we did not investigate the effects of Fasxiator on the activation/cleavage of FXI by FXIIa and thrombin. The inhibition kinetics of rFasxiator on FXIa using chromogenic substrate S2366 (Fig. 3C,D) indicated that rFasxiator is a slow-type inhibitor, with IC_{50} values changing from 3 μmol L^{-1} (no preincubation) to 1.5 μmol L^{-1} (1-h preincubation). In contrast, rFasxiator was a fast-type inhibitor of chymotrypsin with an IC_{50} of 1 μmol L^{-1} (data not shown).

rFasxiator prolongs APTT through inhibition of FXIa

As rFasxiator inhibits the cleavage of small peptide substrates by FXIa, we examined the interaction of rFasxiator with FXIa.

First, we determined the affinity of rFasxiator to FXIa using surface plasmon resonance (Fig. 4A). The sensorgrams indicated strong interaction of rFasxiator with FXIa; we calculated a K_D of 20.2 nmol L^{-1} assuming a 1:1 stoichiometry and single step binding. Similarly,

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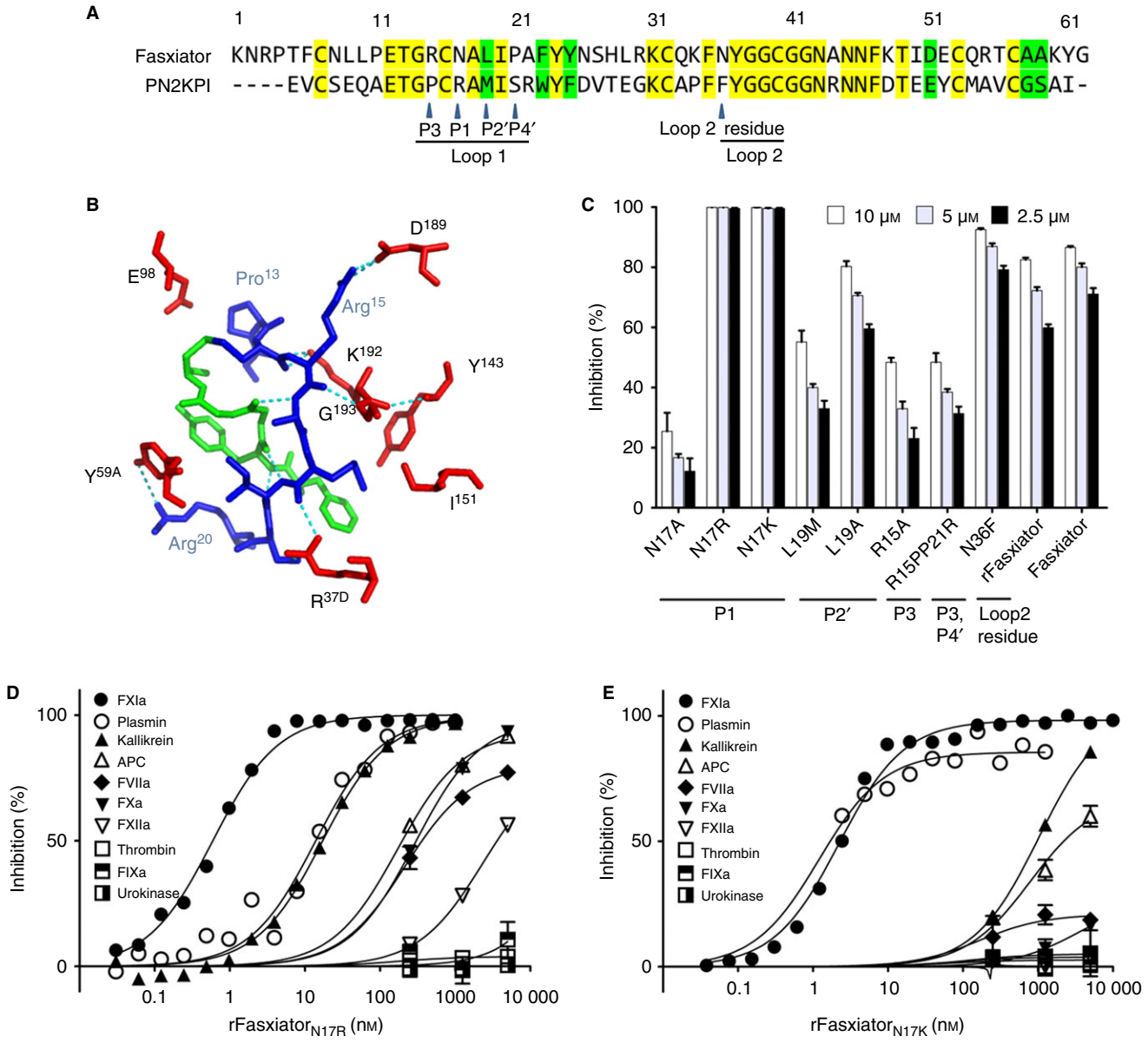


Fig. 5. Structure–function relationships of rFasxiator. (A) Sequence alignment of two FXIa-specific Kunitz-type inhibitors. Yellow color: identical residues; green color: conserved residues. To increase potency and selectivity of rFasxiator, we selected residues (identified by arrowheads) that are involved in interaction with FXIa. (B) Interaction between PN2KPI and FXIa light chain (protease domain), FXIac, as determined by three-dimensional structure of the complex [25]. Only relevant segments are shown. Red: FXIa residues involved in interaction with PN2KPI; dark blue: loop I (GPCRAMISR) residues of PN2KPI; green: loop 2 (FYGGC) residues of PN2KPI; light blue: hydrogen bonds; The guanidinium group of Arg15 (P1) in PN2KPI interacts with the carboxylate group of Asp189 of FXIac to form a salt bridge in the S1 pocket; Pro13 (P3) helps Cys14 (P2) and Gly12 (P4) form hydrogen bonds with Lys192 of FXIac by turning the backbone; Met17 (P2′) and Ser19 (P4′) interact with Arg37D of FXIa; Phe34 of PN2KPI is in van der Waals contact with Tyr143 of FXIac. (C) FXIa inhibition by site-directed mutants of rFasxiator (40-min preincubation with FXIa). Mutations were made through PCRs using pET28a plasmid containing the original Fasxiator gene as a template and primers containing designed mismatch base pair(s). Obtained clones were sequenced and plasmids containing desired mutations were transformed into BL21 competent cells for protein expression. The induction and purification procedures were the same as for rFasxiator. The homogeneity and identity of each mutant were ascertained by ESI-MS (Table S2). Two mutants rFasxiator$_{N17R}$ and rFasxiator$_{N17K}$ show significant increase in their potency; Selectivity profiles of rFasxiator$_{N17K}$ (D) and rFasxiator$_{N17R}$ (E) (30-min preincubation of enzyme and inhibitor). Further mutations are needed for improving selectivity.

rFasxiator interacted with chymotrypsin with a K_D of 29.3 nmol L^{-1} (Fig. S4). This K_D value is similar to that of chymotrypsin inhibitor BFIX (~ 50 nmol L^{-1}) reported earlier [21].

Second, we examined the effect of rFasxiator on the cleavage of FIX, the physiological substrate of FXIa. The cleavage of FIX by FXIa was monitored using Western blotting in both the presence and absence of 10 μmol L^{-1}

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Fig. 6. Functional characterization of rFasxiator$_{N17R,L19E}$. (A) Comparison of selectivity between rFasxiator$_{N17R,L19E}$ and rFasxiator$_{N17R}$. rFasxiator$_{N17R,L19E}$ has better selectivity over FVIIa, FXa, FXIIa, kallikrein, APC, plasmin, and chymotrypsin. (B) Dose-dependent inhibition of FXIa by rFasxiator$_{N17R,L19E}$, rFasxiator$_{N17R}$, and rFasxiator$_{N17K}$ (40-min preincubation). (C) Time-dependent inhibition of FXIa by rFasxiator$_{N17R,L19E}$, rFasxiator$_{N17R}$, and rFasxiator$_{N17K}$. (D) Effect of rFasxiator$_{N17R,L19E}$ on cleavage of S2366 by FXIa. The curves were fitted into Eqn. 1 to determine K_{obs}. (E) K_{obs} plotted against substrate concentration. The decrease in inhibition with increasing concentrations of substrate suggests that rFasxiator$_{N17R,L19E}$ is a competitive inhibitor. (F) K_{obs} values were plotted against rFasxiator$_{N17R,L19E}$ concentration. A straight line indicates that rFasxiator$_{N17R,L19E}$ is a type I slow-type inhibitor. (G) Selectivity of rFasxiator$_{N17R,L19E}$ against different serine proteases (40-min preincubation). rFasxiator$_{N17R,L19E}$ has high potency and good selectivity to FXIa.

rFasxiator. The presence of rFasxiator slowed down the cleavage of FIX (Fig. 4B, upper panel). At 15 min, rFasxiator exerted a dose-dependent inhibition of FIX cleavage by FXIa (Fig. 4B, lower panel). The amount of cleavage product, FIXa, was also quantified by chromogenic assays (Fig. 4C). rFasxiator was found to inhibit the activation of FIX with a similar potency as the inhibition of FXIa cleavage of small peptide substrates (IC_{50} ~ 3 μmol L^{-1}).

Finally, we tested the effect of rFasxiator on the prolongation of APTT in normal and FXI-deficient human plasma. rFasxiator failed to prolong APTT in FXI-deficient plasma, while it was able to prolong APTT in FXI-rescued plasma with a similar potency as in normal human plasma (Fig. 4D). This result strongly suggests that FXI is necessary for rFasxiator's function as an anticoagulant and prolongation of APTT was achieved via FXIa inhibition.

Improvement of rFasxiator potency by site-directed mutagenesis

As the *in vitro* studies showed that rFasxiator prolonged APTT through inhibition of FXIa, the anticoagulant effect of rFasxiator was tested in mouse plasma. Unfortunately, unlike in human plasma, rFasxiator failed to prolong APTT (Fig. S5). This is probably due to the weaker inhibition of murine FXIa.

Compared with PN2KPI, the inhibitory effects of rFasxiator on FXIa are relatively weak. To improve the potency and selectivity of rFasxiator, specific residues in loops 1 and 2 of rFasxiator were selected based on the co-crystal structure [25] of PN2KPI and FXIa catalytic domain (PDB:1ZJD) (Fig. 5A,B). The following single point mutations were carried out: (1) P1 substitutions N17A, N17K, and N17R; (2) P2′ substitutions L19A and L19M; (3) P3′ substitution R15A; (4) P3, P4′ exchange R15P,P21R (all in loop 1), and (5) N36F substitution (in loop 2).

All mutants were tested for FXIa inhibition (Fig. 5C). Mutation of P1, as expected, had a significant effect on the inhibitory activity: rFasxiator$_{N17A}$ lost ~ 75% of the inhibitory activity, while mutation to positively charged Lys and Arg significantly enhanced it. All the other substitutions at P2′, P3′, P4′, and P3 resulted in decreased or had no significant effect on the inhibitory activity. We determined the specificity of rFasxiator$_{N17K}$ and rFasxiator$_{N17R}$ for the serine proteases in the coagulation pathway. Although these mutations increased the potency, the selectivity was lost: rFasxiator$_{N17K}$ has similar selectivity toward FXIa and plasmin (Fig. 5D), while rFasxiator$_{N17R}$ is only marginally selective toward FXIa compared with plasmin and kallikrein (Fig. 5E).

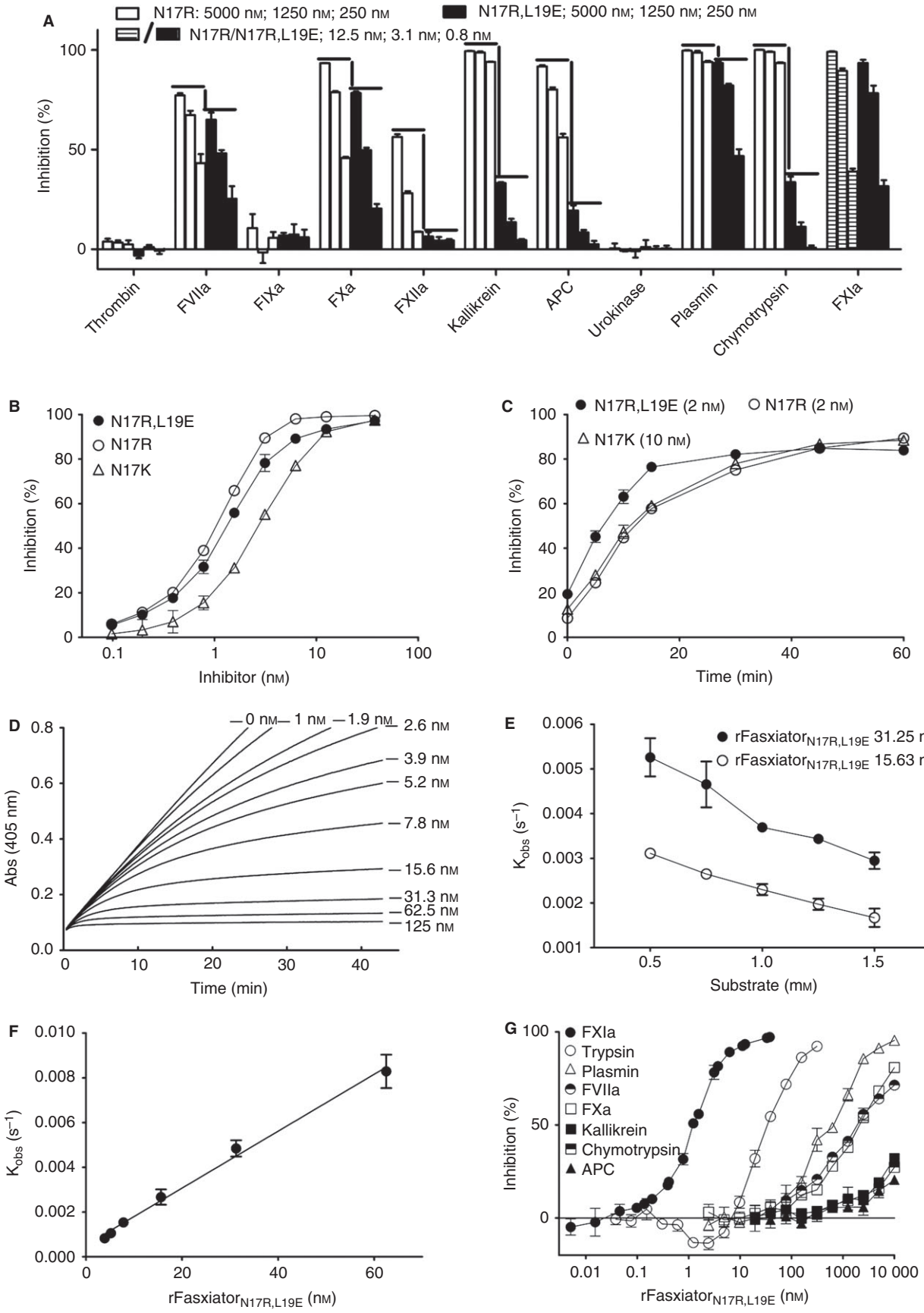
To enhance the selectivity of rFasxiator, we selected rFasxiator$_{N17R}$ for further modifications. The P2′ L residue was mutated to A/E/F/N/R/S. All the double mutants were then evaluated for their ability to inhibit FXIa and plasmin (Fig. S6). Glutamate in P2′ position is the least preferred residue by kallikrein, APC, plasmin, FXIIa, and chymotrypsin (Fig. S7); rFasxiator$_{N17R,L19E}$ showed dramatic loss of inhibitory effect on these enzymes (Fig. 6A). In contrast, FXIa tolerated an acidic residue in P2′ position. This glutamate may interact with one of the several positively charged residues of FXIac and form a salt bridge. Interestingly, kallikrein (PDB: 2ANY), despite high similarity to FXIa, shows 6000-fold lower affinity. Crystal structures of the complexes with FXIa and kallikrein may help describe the subtleties of the interactions. The dose-response studies indicated that the IC_{50} values for FXIa inhibition by rFasxiator$_{N17R}$, rFasxiator$_{N17K}$, and rFasxiator$_{N17R,L19E}$ with 40-min incubation were 0.94, 2.96, and 1.26 nmol L^{-1}, respectively (Fig. 6B). The inhibitory potencies of these mutants increased with prolongation of incubation time with FXIa (Fig. 6C) and hence are also slow-type inhibitors. Thus, with two mutations in rFasxiator$_{N17R,L19E}$, we have significantly improved the inhibitory potency (~ 1000-fold) and selectivity toward FXIa.

Inhibition kinetics of rFasxiator$_{N17R,L19E}$

To further understand the interaction with FXIa, we determined the kinetic parameters of rFasxiator$_{N17R,L19E}$. Progress curves were plotted of S2366 cleavage by FXIa with (i) fixed substrate concentration and different concentrations of rFasxiator$_{N17R,L19E}$ (Fig. 6D) and (ii) fixed rFasxiator$_{N17R,L19E}$ concentration and different concentrations of S2366 (Fig. S8). The curves were fitted into Eqn. 1 to determine K_{obs} values. K_{obs} decreases with the increasing substrate concentration (Fig. 6E). Hence, rFasxiator$_{N17R,L19E}$ is a competitive inhibitor. As the plot of K_{obs} vs. rFasxiator$_{N17R,L19E}$ concentrations (Fig. 6F) shows a straight line, rFasxiator$_{N17R,L19E}$ is a type I slow-type inhibitor [16] with a K_i of 0.86 nmol L^{-1} using Eqns. 2 and 3. For comparison, we have also determined the dose-

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Table 1 Comparison of K_i of rFasxiator$_{N17R,L19E}$ with PN2KPI

	rFasxiator$_{N17R,L19E}$/-fold to FXIa	PN2KPI/-fold to FXIa
FXIa	0.86 nmol L^{-1}/1	0.81 nmol L^{-1}/1
Trypsin	9.7 nmol L^{-1}/11.3	0.03 nmol L^{-1}/−27
Kallikrein	5297.7 nmol L^{-1}/6160.1	183 nmol L^{-1}/225.9
Plasmin	145.5 nmol L^{-1}/169.2	459 nmol L^{-1}/566.7
FVIIa	693 nmol L^{-1}/805.8	3500 nmol L^{-1}/4320
FXa	718 nmol L^{-1}/834	1150 nmol L^{-1}/1419.8

dependent inhibition of other serine proteases (Fig. 6G). K_i values of rFasxiator$_{N17R,L19E}$ inhibiting of trypsin, plasmin, FXa, FVIIa, and kallikrein were obtained and compared with those of PN2KPI (Table 1). rFasxiator$_{N17R,L19E}$ inhibited FXIa with similar potency as PN2KPI (0.86 vs. 0.81 nmol L^{-1}), while their selectivities are different: compared with PN2KPI, rFasxiator$_{N17R,L19E}$ dropped 2- to 5-fold in selectivity against plasmin, FXa, and FVIIa but gained 28.9- and 323-fold selectivity against kallikrein and trypsin. Thus, we have developed a new specific inhibitor of FXIa, which has similar potency as its endogenous physiological inhibitor but different selectivity.

rFasxiator$_{N17R,L19E}$ prolongs $FeCl_3$-induced carotid artery thrombosis

The anticoagulant effects of rFasxiator$_{N17R,L19E}$ were quantified by APTT and PT in human and murine plasma. rFasxiator$_{N17R,L19E}$ prolonged APTT in a dose-dependent manner (Fig. 7A) and doubled APTT at ~ 300 nmol L^{-1}, but had no significant effect on PT up to 40 μmol L^{-1} in human plasma (Fig. S9). Similarly, it also prolonged APTT in mice plasma in a dose-dependent manner; however, the potency was ~ 10-fold weaker (doubled APTT at ~ 4 μmol L^{-1}). Thus, rFasxiator$_{N17R,L19E}$ exerts its anticoagulant effects through the intrinsic pathway.

We used an established $FeCl_3$-induced carotid artery thrombosis model to evaluate the *in vivo* antithrombotic efficacy of rFasxiator$_{N17R,L19E}$. In the first set of experiments, 0.2 mg/mice rFasxiator$_{N17R,L19E}$ prolonged the occlusion time by ≈ 2-fold (5.57 vs. 9.55 min) in the test animals compared with control animals (Fig. 7B). While all the control animals conformed to a one-step formation of thrombus (flow gradually reduced to zero and maintained for 30 min), four of the treated animals showed diffused events (flow reduced first, reverted back to normal, and reduced again). Therefore, the animal experiments were repeated with the increased dosage of inhibitor (0.3 mg/mice) and reduced injury strength. Under these conditions, six control animals had their occlusion times prolonged to 10 min, but no occlusion was found until 30 min in the seventh mouse. In treated animals, there was no occlusion until the end of experimental period (30 min) (Fig. 7C). Thus, rFasxiator$_{N17R,L19E}$ exhibited antithrombotic effects in mouse carotid artery thrombosis. The *in vivo* data complemented with our *in vitro* data demonstrate the potent anticoagulant activity of rFasxiator$_{N17R,L19E}$.

Discussion

Fasxiator described in this work is an exogenous specific FXIa inhibitor identified from snake venom using assay-guided purification of anticoagulants targeting the intrinsic coagulation pathway. It is a Kunitz-type protease inhibitor that has distinct amino acid sequence compared with other reported trypsin, chymotrypsin, and FXIa inhibitors. The properties of rFasxiator was improved through site-specific mutagenesis, and the anticoagulant effects of a double mutant, rFasxiator$_{N17R,L19E}$, which possesses both high potency and selectivity, was confirmed in an established animal model.

rFasxiator$_{N17R,L19E}$-treated animals were protected against $FeCl_3$-induced carotid artery thrombosis, and the protective effects are related to the intensity of vascular injury (Fig. 7), which is similar to the observations made with FXI-deficient mice [20]: FXI-deficient mice were fully protected against thrombosis induced by 5% $FeCl_3$, partially protected against 7.5% $FeCl_3$, and showed almost no protection against 10% $FeCl_3$. Further, the diffused pattern (flow reduced first, reverted back to normal, and reduced again) observed in several rFasxiator$_{N17R,L19E}$-treated animals suggested that rFasxiator$_{N17R,L19E}$ played a role in the growth/stability of thrombus, which is consistent with the phenomena observed elsewhere in FXI/FXIa disrupted/inhibited animals: platelet deposition and fibrin accumulation were reduced in vascular graft occlusion in FXI antibody treated–baboon [26], and thrombi inside the carotid artery was flushed away by blood flow in some of the PN2KPI-treated mice [18]. These observations support that rFasxiator$_{N17R,L19E}$ targets FXIa *in vivo*.

There are several approaches used to target FXI/FXIa for therapeutic purposes [27,28]. The first is mouse anti-FXI monoclonal antibodies [26]; although a number of monoclonal antibodies are in therapeutic use, in general they are recognized as expensive [29]. Second, to alleviate, potential allergic reactions such antibodies have to be humanized [30]: antisense oligonucleotides [31,32]; since FXI has a half-life of 52 h in human plasma, they have a delayed effect. Third, synthetic active site/allosteric inhibitors [33–35] typically exhibit potent *in vitro* activity

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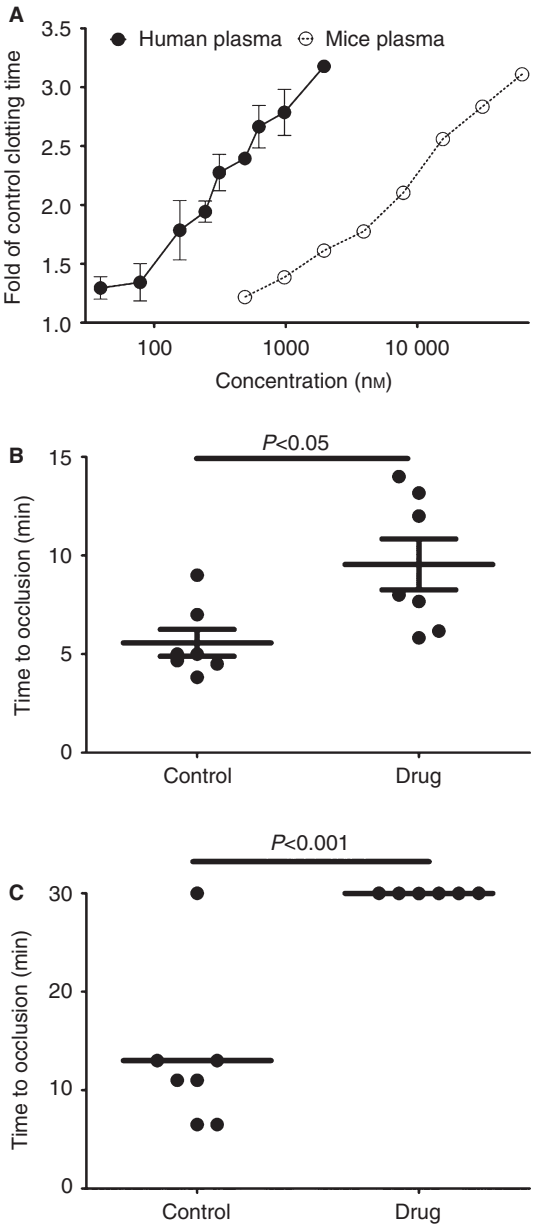


Fig. 7. Antithrombotic effect of rFasxiator$_{N17R,L19E}$ in $FeCl_3$-induced carotid artery thrombosis model in mice. (A) Potency of rFasxiator in prolonging APTT in both human and murine plasma. Human plasma data represent average of triplicates, while mice plasma data represent average of duplicates. (B) Thrombosis was induced by 6% $FeCl_3$, 2 mm × 2 mm filter paper; 200 μg rFasxiator$_{N17R,L19E}$ per mouse (seven mice per group). Significant differences between two groups were calculated using *t*-test, $P < 0.05$. (C) Thrombosis was induced by 5% $FeCl_3$, 1 mm × 1 mm filter paper; 300 μg rFasxiator$_{N17R,L19E}$ per mouse (six or seven mice per group). Significant differences between two groups were calculated using *t*-test, $P < 0.001$.

(0.1–1 nmol L^{-1}) in enzymatic assays but have a > 1000-fold drop in potency when tested for APTT (> 1 μmol L^{-1}) with human plasma. Further, they have poor selectivity (< 30-fold specificity). Fourth are naturally occurring inhibitors, which include Clavatadine A, Desmolaris, PN2KPI, and Fasxiator [36,37]. Clavatadine A has poor potency (IC_{50} = 1.3 μmol L^{-1}), while Desmolaris has poor selectivity (K_i for FXIa = 12.35 nmol L^{-1}, K_i for FXa = 15.06 nmol L^{-1}). It is unclear whether the high affinity inhibition of FXIa and FXa by Desmolaris may have an added advantage as an anticoagulant. Both PN2KPI and Fasxiator are Kunitz-type inhibitors with high specificity and selectivity. These quick-acting (within 30 min) inhibitors are much smaller and can be produced and purified cost-efficiently compared with antibodies. Similar small, less antigenic, disulfide-rich proteins, such as aprotonin and hirudin, are being used safely as therapeutic agents. Their limited similarity (58%) in amino acid sequence suggests different stabilities, physiological, and pharmaceutical properties. For example, their selectivities are different: compared with PN2KPI, rFasxiator$_{N17R,L19E}$ dropped 2- to 5-fold in selectivity against plasmin, FXa and FVIIa, but gained 28.9- and 323-fold selectivity against kallikrein and trypsin. While rFasxiator$_{N17R,L19E}$ was successfully expressed using *E. coli*, PN2KPI (produced using yeast) cannot be expressed under similar conditions in our laboratory. Endogenous PN2KPI may be less antigenic compared with Fasxiator. However, if some patients develop immune response, PN2KPI may lead to severe autoimmune side effects even after its use is discontinued.

Kunitz-type inhibitors are widely reported in Viperidae and Elapidae snake venoms [38]. With the exception of dendrotoxin (potassium channel blocker) [39] and calcicludine (calcium channel blocker) [40], all other proteins are known to inhibit various proteases. To evaluate acute toxicity, we administered rFasxiator or rFasxiator$_{N17R,L19E}$ at a single dose of 50 mg kg^{-1} (intraperitoneal) or 10 mg kg^{-1} (intravenous) injections into C57BL/6 mice. These mice showed no pathologic symptoms, and they survived 3 days without any complications (animals were not observed > 3 days). Although further studies are needed to evaluate toxicity, rFasxiator and rFasxiator$_{N17R,L19E}$ do not have severe high toxicity expected for snake toxins. Thus, rFasxiator and its variants may provide new options for developing agents targeting FXI/FXIa for therapeutic purpose.

In summary, we have identified and characterized a specific FXIa inhibitor from the venom of *B. fasciatus,* engineered it to improve its potency by ~ 1000 times, and demonstrated its *in vitro* and *in vivo* efficacy. The proof-of-principle data reported here support the further development of Fasxiator as a novel anticoagulant candidate.

Addendum

W. Chen designed and performed experiments, analyzed results, and wrote the manuscript. L. P. D. Carvalho and M. Y. Chan performed the $FeCl_3$-induced carotid artery thrombosis model experiments. R. M. Kini designed experiments and wrote and reviewed the manuscript. T.

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S. Kang designed and supervised experiments and wrote the manuscript.

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Disclosure of Conflict of Interests

W. Chen, T. S. Kang, and R. M. Kini have a 'Compositions and methods for inhibiting thrombogenesis' patent pending.

Supporting Information

Additional Supporting Information may be found in the online version of this article:

Fig. S1. Serine proteases specificity of BF01 and BF02.
Fig. S2. Sequence determination of BF01/02.
Fig. S3. Effects of rFasxiator on the intrinsic and the extrinsic tenase complexes.
Fig. S4. High affinity binding of rFasxiator to chymotrypsin.
Fig. S5. Effects of rFasxiator on APTT of human (A) and murine (B) plasma.
Fig. S6. Selectivity of double variants part 1.
Fig. S7. Selectivity of double variants part 2.
Fig. S8. Effects of substrate (S2366) concentrations on the inhibition of FXIa by $rFasxiator_{N17R,L19E}$.
Fig. S9. Effects of $rFasxiator_{N17R,L19E}$ on PT assays in human plasma.
Table S1. Molecular weights of proteins in RP-HPLC pooled fractions.
Table S2. Molecular weights of rFasxiator mutants.

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